Bus-Level Inertia Distribution Analysis in IEEE 39-Bus System

— Under Increasing Renewable Energy Penetration —

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1. Introduction

Why Bus Inertia Matters

Research Background

The Low-Inertia Challenge

- Renewable energy sources (wind, solar) are replacing synchronous generators
- Inverter-based resources (IBRs) provide **zero rotational inertia**
- System frequency becomes more vulnerable to disturbances

Key Question

> **Where** in the network is inertia support weakest, and **how much** renewable penetration can the system tolerate before frequency stability is compromised?

System Inertia Fundamentals

$$H_{\text{sys}} = \frac{\sum_{i=1}^{N_g} H_i \cdot S_i}{\sum_{i=1}^{N_g} S_i}$$

Symbol	Description

Rate of Change of Frequency (RoCoF)

$$\mathrm{RoCoF} = \left.\frac{df}{dt}\right|_{t=0^+} = -\frac{\Delta P_{\mathrm{MW}} \cdot f_0}{2 \cdot \underbrace{\sum_i H_i \cdot S_i}_{E_{\mathrm{kin}}}}$$

Symbol	Description
ΔP_{MW}	Power imbalance (MW) — e.g., generator trip
f_0	Nominal frequency (60 Hz)

RoCoF is inversely proportional to system kinetic energy. Less inertia → faster frequency decline.

2. Methodology

Bus-Level Inertia Distribution

Electrical Distance

$$d_{ij} = |Z_{ii} + Z_{jj} - 2Z_{ij}|$$

Symbol	Description
$\mathbf{Z}_{\text{bus}}$	Bus impedance matrix: $\mathbf{Z}_{\text{bus}} = \mathbf{Y}_{\text{bus}}^{-1}$
d_{ij}	Electrical distance between bus *i* and bus *j*

Electrical distance captures how "close" a bus is to a generator in terms of network impedance.

Effective Inertia at Each Bus

$$H_{\text{eff}}(k) = \frac{\sum_{i=1}^{N_g} w_{ki} \cdot H_i \cdot S_i}{S_{\text{base}}}$$

$$w_{ki} = \frac{1/d_{ki}}{\sum_{j=1}^{N_g} 1/d_{kj}}$$

Symbol	Description

A generator's inertia contribution is stronger at electrically closer buses.

Aggregated Swing Equation

$$2H_{\text{sys}} \frac{d\Delta f}{dt} = -\frac{\Delta P}{S_{\text{total}}} - D \cdot \Delta f + P_{\text{gov}}$$

$$T_{\text{gov}} \frac{dP_{\text{gov}}}{dt} = -\frac{\Delta f}{R} - P_{\text{gov}}$$

Parameter	Value	Description
D	2.0 pu	System damping coefficient
T_{gov}	0.2 s	Governor time constant
R	0.05 pu	Droop setting

3. Test System

IEEE 39-Bus New England System

IEEE 39-Bus Generator Parameters

Table 1. Generator data (10 synchronous machines, 7,700 MVA total)

30	G1	4.20	350	1,470
31	G2	3.03	830	2,515
32	G3	3.58	620	2,220
33	G4	2.86	750	2,145
34	G5	2.60	560	1,456
35	G6	3.48	680	2,366
36	G7	2.64	580	1,531
37	G8	2.43	890	2,163
38	G9	3.45	1,040	3,588

$H_{\mathrm{sys}} = 3.436$ s, $E_{\mathrm{kin}} = 26{,}454$ MWs. Replacement order: G8→G5→G7→G4→G2→G3→G1→G6 (ascending H).

4. Results

Base Case & Frequency Response

Network Topology with Inertia Overlay

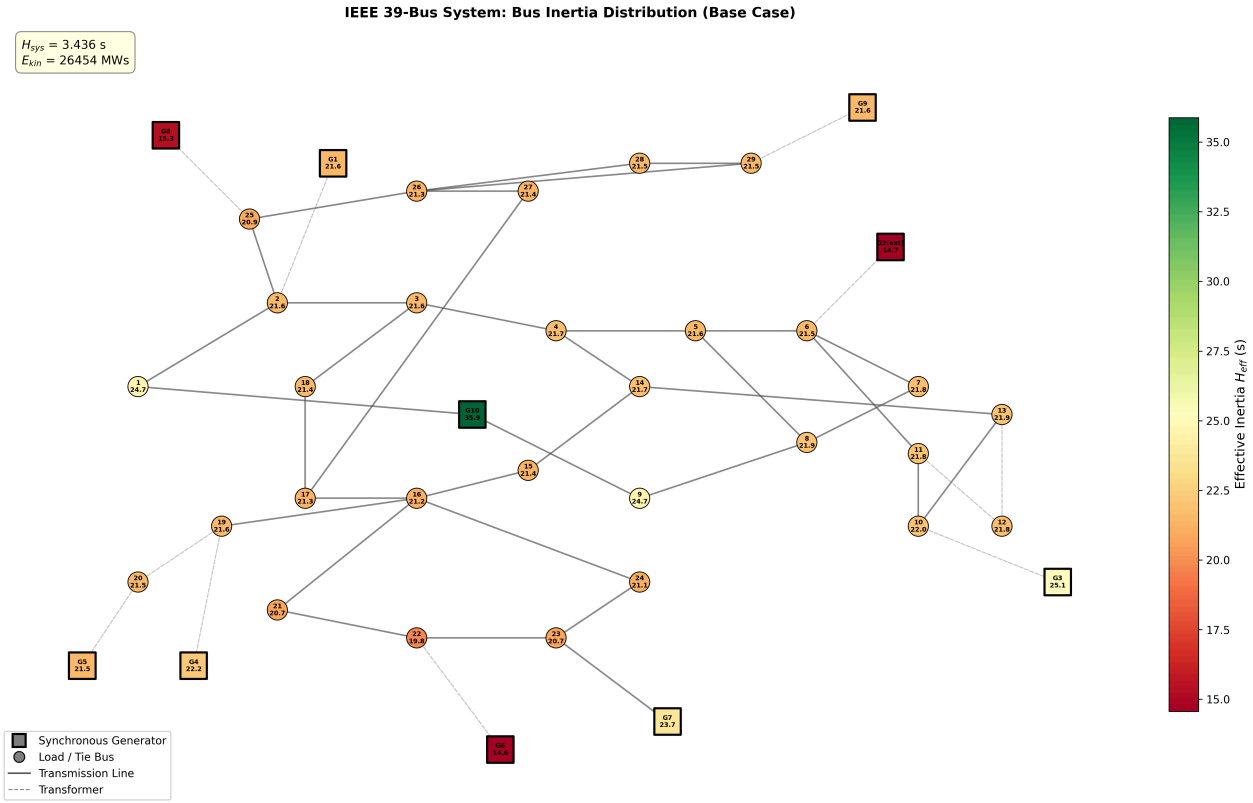


Fig. 1. IEEE 39-bus system topology. Squares = generator buses, circles = load buses. Color: green (high H_{eff}) → red (low H_{eff}). Bus 38 has the highest inertia (35.88 s), Bus 34 the lowest (14.56 s).

Inertia-Weak Bus Identification

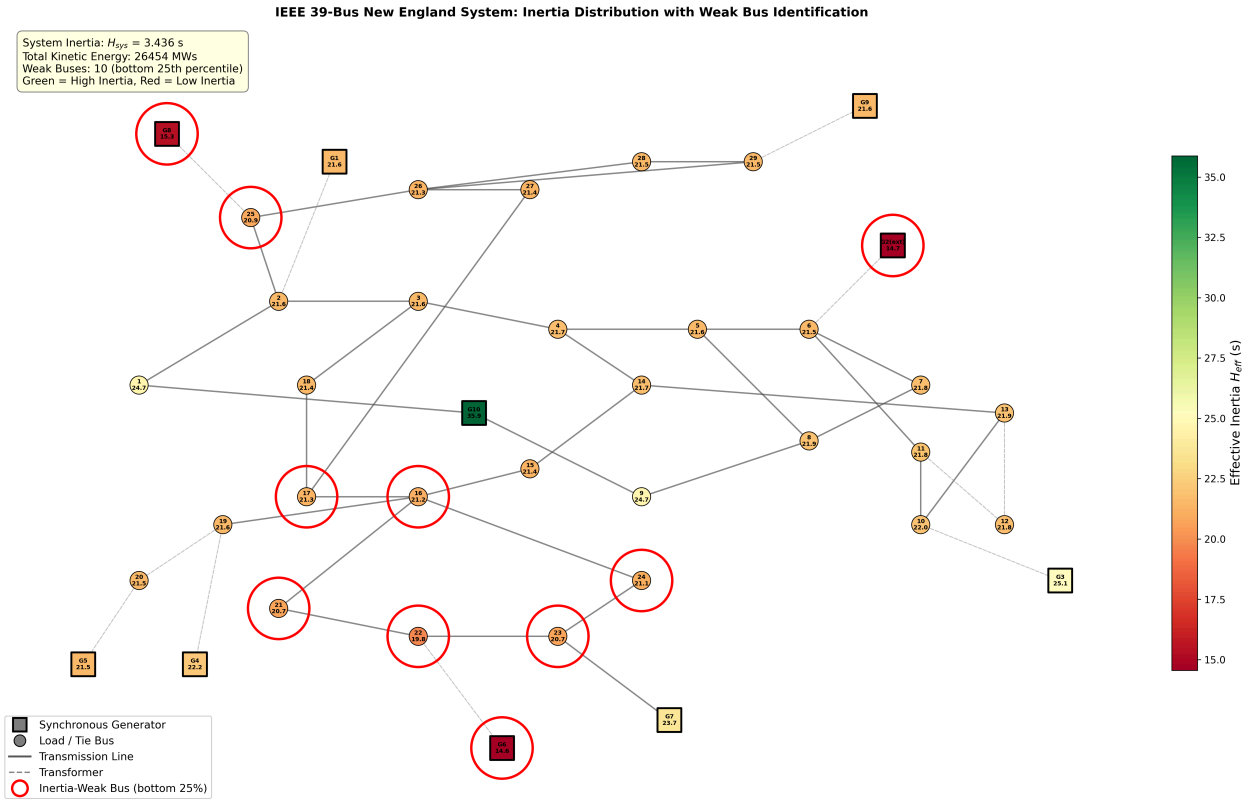


Fig. 2. Red circles highlight 10 inertia-weak buses (bottom 25th percentile). Two clusters: (1) southern network (Bus 20–24), (2) small generator buses (G1, G5, G7).

Bus Inertia Distribution (Bar Chart)

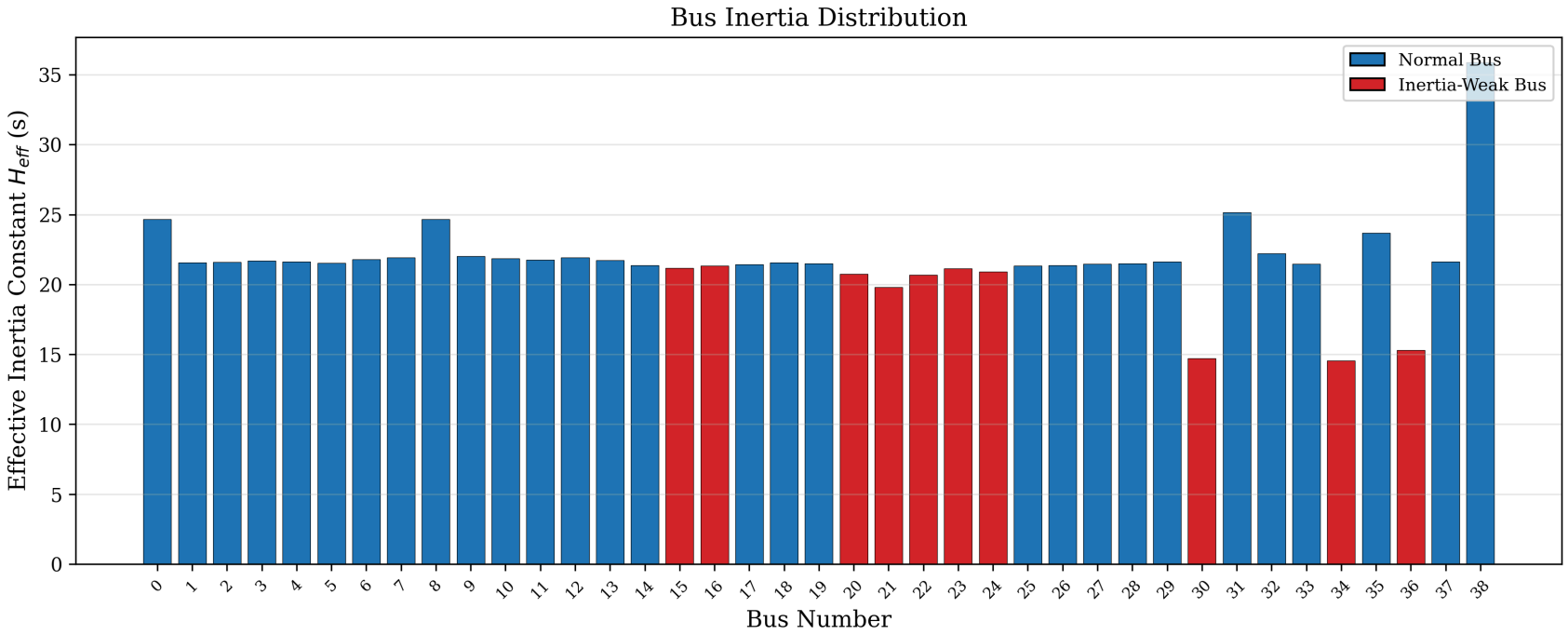


Fig. 3. Effective inertia H_{eff} at each bus. Red bars = inertia-weak buses. A 2.5:1 ratio exists between highest and lowest inertia buses.

Frequency Response — G10 Trip (1,000 MW)

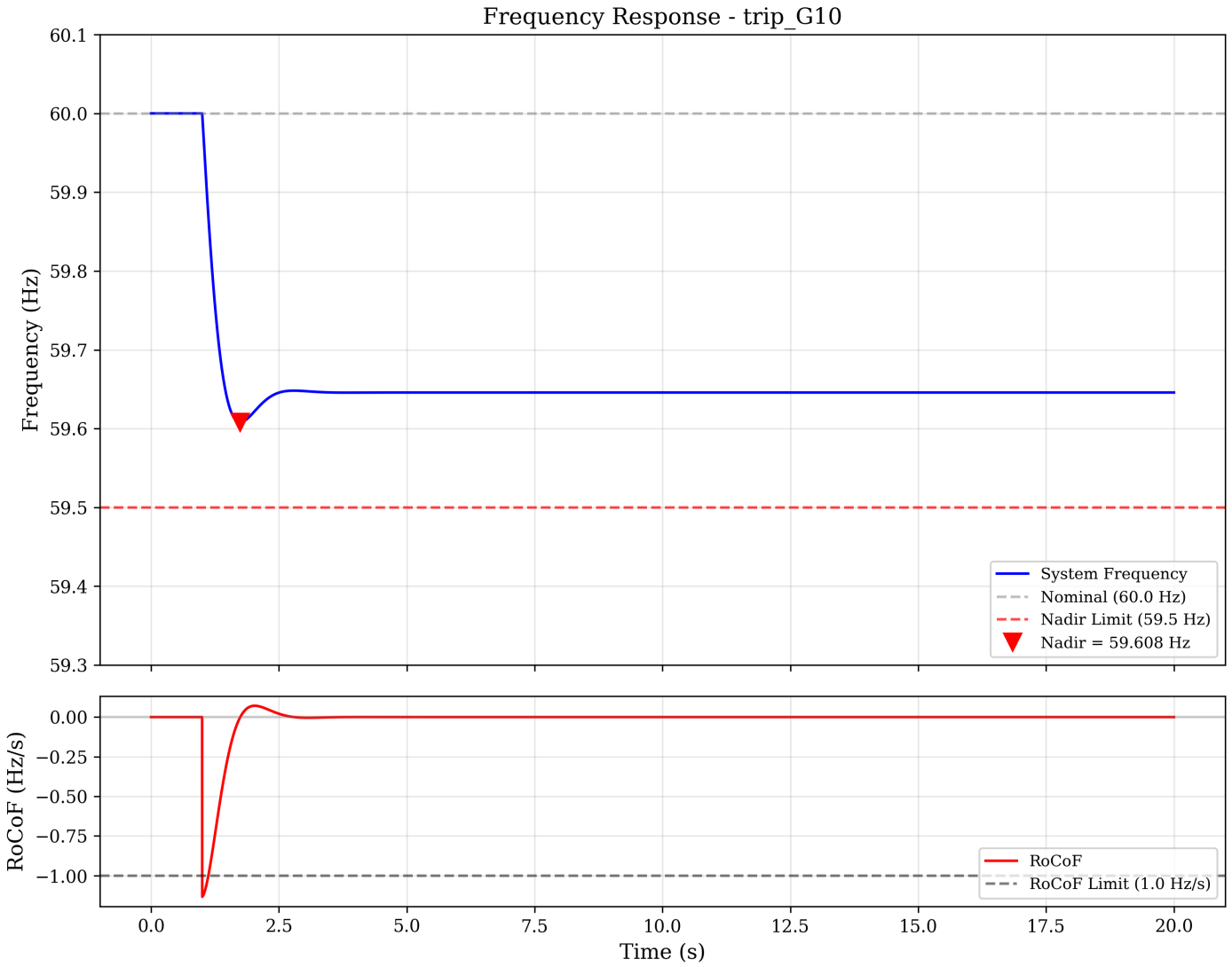


Fig. 4. Top: frequency trajectory (nadir = 59.608 Hz). Bottom: RoCoF = −1.134 Hz/s, exceeding the 1.0 Hz/s grid code limit.

Generator Trip Scenarios

Table 2. Frequency stability metrics for three disturbance scenarios

Trip G8	540	–0.612	59.788	✓	✓
Trip G10	**1,000**	**–1.134**	**59.608**	**✗**	✓
Trip G2	573	–0.650	59.776	✓	✓

Only the largest generator trip (G10, 1,000 MW) violates the RoCoF limit. Analytical and simulated RoCoF agree within **0.2%**.

5. Renewable Penetration Impact

From 0% to 80%

Impact on RoCoF, Nadir, and H_{sys}

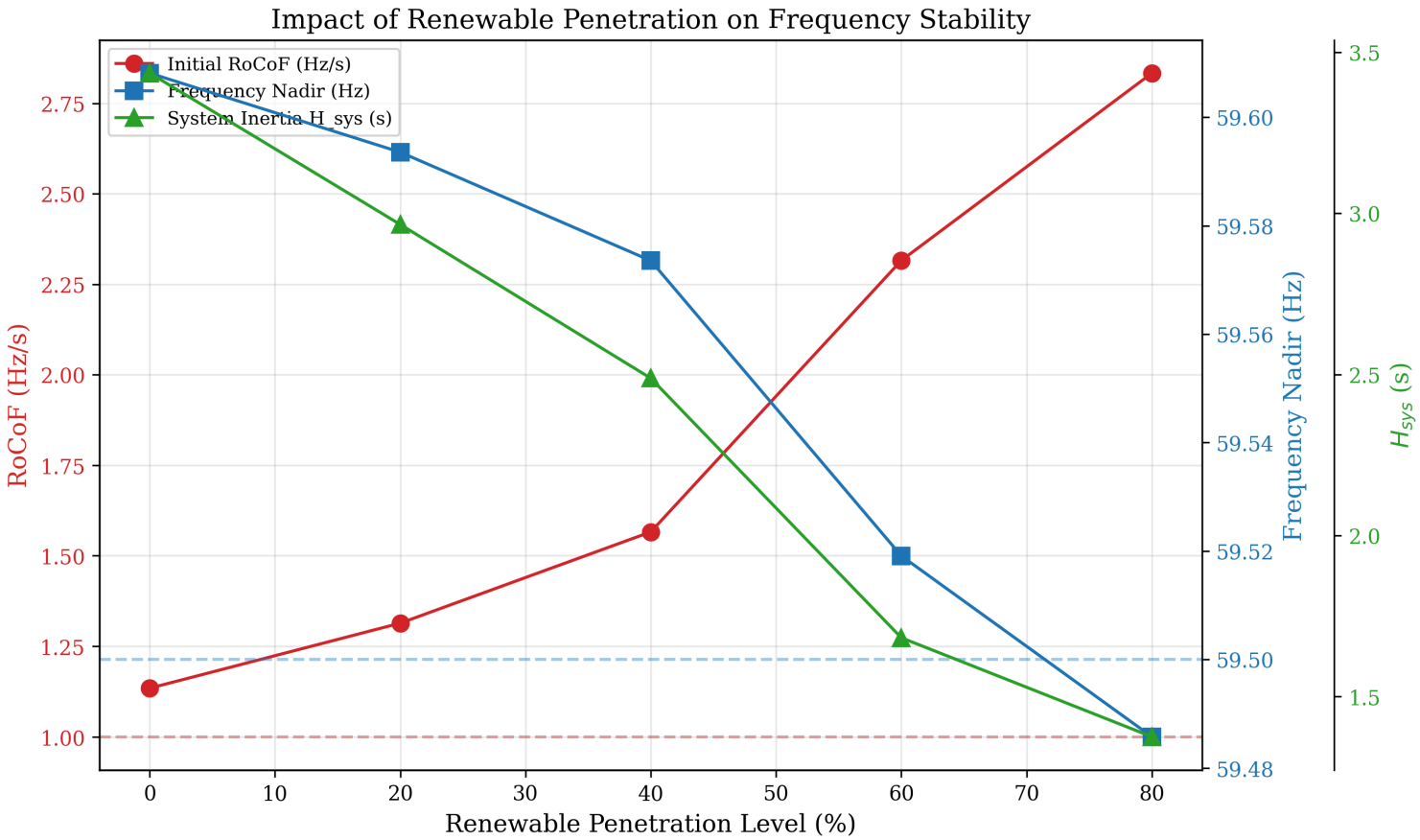


Fig. 5. RoCoF worsens from –1.13 to –2.83 Hz/s (2.5× degradation). Frequency nadir violates 59.5 Hz limit at 80% penetration. Dashed lines = grid code thresholds.

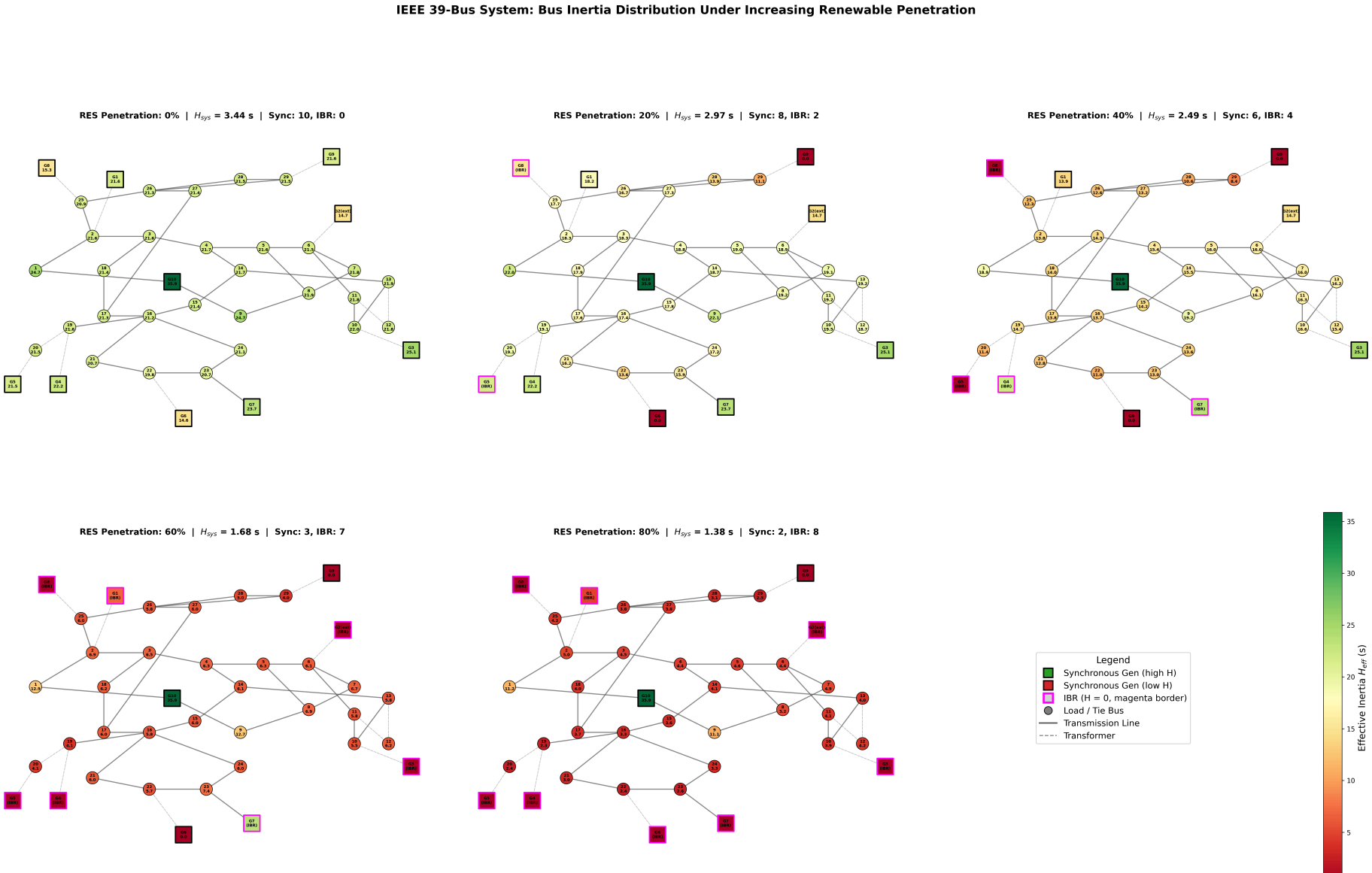
Penetration Study — Detailed Results

Table 3. G10 trip (1,000 MW) under increasing RES penetration

0	3.436	26,454	−1.134	59.608	—
20	2.966	22,835	−1.314	59.594	G8, G5
40	2.488	19,159	−1.566	59.574	+G7, G4
60	1.682	12,954	−2.316	59.519	+G2, G3, G1
80	**1.375**	**10,588**	**−2.833**	**59.486**	+G6

60% reduction in H_{sys}. Critical threshold: **60–80%** penetration.

Spatial Inertia Under Penetration (Topology)



Bus-Level Inertia & Multi-Machine Dynamics

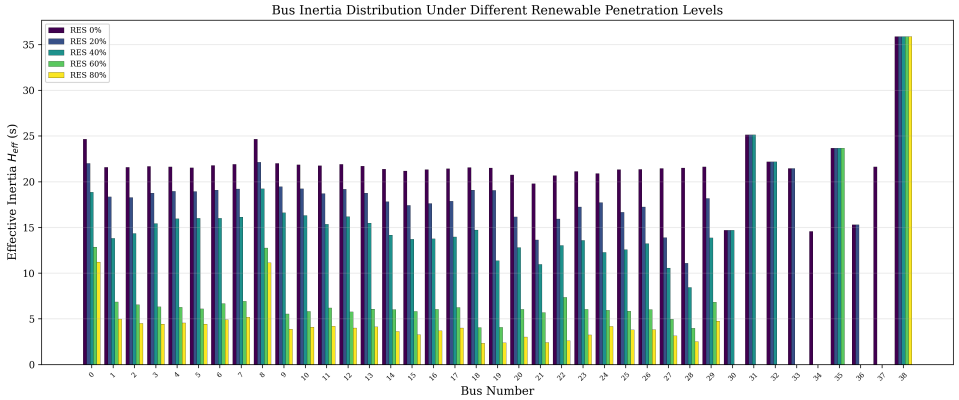


Fig. 7. H_{eff} comparison across 5 penetration levels. Inertia uniformly degrades except near G9/G10.

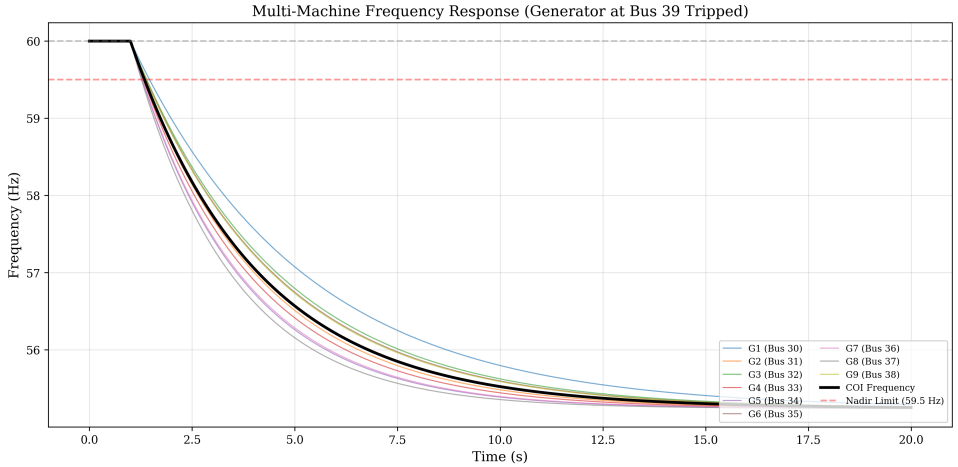


Fig. 8. Multi-machine response (0% RES). Low-H generators (G5: 2.60 s) deviate faster than high-H machines (G9: 3.45 s).

6. Key Findings

Summary of Key Findings

Comprehensive bus-level inertia analysis reveals significant spatial heterogeneity and identifies the critical renewable penetration threshold.

Spatial Distribution

1. **2.5:1 ratio** between highest (Bus 38) and lowest (Bus 34) inertia
2. **10 weak buses** cluster in two regions
3. At 80% RES, inertia concentrated in only **2 locations**

Frequency Stability

1. H_{sys} drops **60%** (3.44→1.38 s)
2. RoCoF worsens **2.5×** (–1.13→–2.83 Hz/s)
3. **Critical threshold: 60–80%** penetration
4. Nadir violates 59.5 Hz at 80%

Conclusion: Bus-level inertia analysis is essential for identifying spatially vulnerable regions. Virtual inertia support should be prioritized at inertia-weak buses (Bus 20–24, G1, G5, G7) to maintain frequency stability under high RES penetration.

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Thank you

Questions?

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